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Double-image encryption based on spatiotemporal chaos and DNA operations

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Abstract
A double-image encryption algorithm combining DNA insertion and deletion operations with spatiotemporal chaos is proposed. The technology of DNA sequence insertion and deletion is employed in both confusion and diffusion to encrypt two images simultaneously. In the proposed image encryption scheme, the key streams to control DNA operations are obtained by means of iterating the system of non-adjacent coupled map lattices (NCML). The original key streams are related to the plain-image, which increases the sensitivity of the proposed algorithm. Two confused images are served as keys to encrypt each other. The confused image is associated with another through the DNA insertion and deletion, which not only improves the security of image encryption system but also enhances the encryption speed. It is demonstrated that the proposed image encryption algorithm has the characteristics of large key space, high sensitivities of key and plain-image, and high entropy. Also, the proposed algorithm is robust to occlusion-attack and noise-attack. Therefore, the proposed double-image encryption algorithm is heuristic for the DNA computing applications into the multi-image encryption.

Keywords Double-image encryption · DNA insertion and deletion · Spatiotemporal-chaos · DNA-level confusion and diffusion

1 Introduction

With the growing techniques of communication and network, information is transmitted over diverse public networks and stored on various platforms. It is an essential and urgent challenge to improve the security of the network. The image is encrypted

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efficiently in which the key issue of information security research is to keep information security during the process of transmission and preservation [7].

Recently, different encryption algorithms are proposed constantly, such as the image encryption algorithms based on compressive sensing [13, 36], chaos [14, 27, 30, 32], DNA computing [3, 18, 19] and neural network [1, 15]. In the past decade, the chaos-based image encryption attracted extensive concerns owing to the chaotic characteristics of nonlinear and sensitivity to the initial conditions. Many chaos-based image encryption algorithms were put forward and acquire positive effect [11, 16, 20, 24]. However, the digitalization result of chaos is periodic eventually because of the finical computer precision and temporal discretization, which will cause the chaotic cryptosystem susceptible to break up [22]. To overcome the short period issue existing in digitalization of chaos, spatiotemporal chaotic system is applied in chaotic cryptography widely due to its excellent chaotic dynamic characteristics, such as larger parameter range and better randomness [6, 31, 34]. However, the image encryption method only based on spatiotemporal chaotic system is insecure enough [29]. Therefore, it is necessary to combine different mechanisms to enhance the security of the image encryption algorithm [12, 26].

Due to the features of vast parallelism and ultra-low power consumption, DNA computing has entered in the domain of image encryption [4, 33]. The kernel of based DNA method is DNA encoding and DNA computing which include some biological operations and algebra operations on a DNA sequence, for instance, the complementary rule of bases, DNA addition, DNA subtraction, and DNA exclusive-OR operation [4]. A number of image encryption methods were came up with combining DNA computing with chaotic encryption to improve the security level of encryption algorithms [2, 4, 5, 8, 23, 28, 33, 34]. A color image encryption algorithm based on DNA sequence operations and multiple improved 1D chaotic system was put forward, where the sequence generated from the Logistic map was not random [23]. Subsequently, the high-dimensional chaotic systems were utilized to structure the image encryption algorithms because they could alleviate the dynamical degradation and provide multiple chaotic sequences [10, 17]. Unfortunately, these schemes [10, 17] were weak against the chosen-ciphertext and chosen-plaintext attacks [2, 28]. In [21], Wang et al. put forward an encryption method via combining DNA sequence operations with coupled map lattices (CML), in which the extended Hamming distance was used to generate new initial values for the CML iteration to fight against chosen-plaintext attack. Nevertheless, a series of common DNA operations were applied in image encryption algorithms [2, 10, 21] to diffuse the DNA-encoded image with the absence of DNA-level confusion, which made the image encryption algorithms inadequate to resist some attacks. And since the traditional DNA computing methods are essentially arithmetic operations of binary numbers, the image encryption schemes based on DNA computing cannot run on the DNA computer to achieve high encryption/decryption speed.

To overcome the weaknesses above-mentioned and to innovate the DNA computing, a new double-image encryption algorithm based on non-adjacent coupled map lattices and DNA computing is proposed. The key streams that control the length of the deletion sequences and the insertion sequences are generated by the NCML. Furthermore, the key streams are also related to the plain-images, which could enhance the security level and withstand the chosen-plaintext and the known-plaintext attacks. And DNA-sequences re-arrangement operation is performed to the rows and columns of double plaintext images respectively by means of DNA deletion and DNA insertion, which cracks the strong correlations among adjacent image pixels effectively. To strengthen the security of the image encryption algorithm, the same re-arrangement operation of DNA sequences is applied to one DNA matrix with another DNA matrix served as the key. The DNA deletion and insertion is applied to both scrambling and diffusion processes, which not only innovates the DNA encryption methods but also increases the running speed. If image encryption algorithms have higher plaintext sensitivity, they are difficult to resist noise attacks. However, in our proposed scheme, we can strike a balance between plaintext sensitivity and anti-noise attack ability. And to date, DNA-level permutation and DNA-level diffusion are never used for two images simultaneously. Simulations reveal that the viability and superiority of the presented encryption scheme.

The remainder of the paper is outlined as follows. In Section 2, DNA operations and NCML are briefly introduced. In Section 3, the proposed image encryption and decryption methods are explained. Experimental results and performance analyses are provided in Section 4 and Section 5 respectively. Finally, conclusions are drawn in the last Section.

2 Preliminaries

2.1 Non-adjacent coupled map lattices

The spatiotemporal chaotic systems maintain long periodicity in digitalization of the computer and guarantee complex dynamical behavior or pseudo-randomness compared with the simple chaotic systems. It is proved that a lot of image encryption algorithms could enhance the security of the cryptosystem established on the CML [9, 21, 25]. Here, a spatiotemporal chaos system of NCML is applied to our scheme. Different from the adjacent CML, the NCML generates chaotic sequences with better pseudo randomness.

NCML is defined as:

$$x_{n+1}(i) = (1-\varepsilon)f[x_n(i)] + \frac{\varepsilon}{2}\{f[x_n(j)] + f[x_n(k)]\}, \tag{1}$$

where i, j and k represent the space lattices $(1 \le i, j, k \le L)$, the coupling factor $\varepsilon \in [0, 1]$. n is the time variable. $f(x)$ is a first-order difference equation:

$$f(x) = \mu x(1-x), \tag{2}$$

where $\mu \in (0, 4]$. The relations of i, j and k are calculated by a non-adjacent map usually described in non-linear maps such as Arnold cat map, tent map and standard map. The relations of i, j, k in the constructed algorithm are defined by Arnold cat map:

$$\begin{pmatrix} j \\ k \end{pmatrix} = A\begin{pmatrix} i \\ i \end{pmatrix} \bmod L = \begin{pmatrix} 1 & p \\ q & pq+1 \end{pmatrix}\begin{pmatrix} i \\ i \end{pmatrix}, \tag{3}$$

where p and q are mapping parameters.

2.2 DNA coding

A DNA molecule consists of 4 nucleic acid bases A (adenine), G (guanine), C (cytosine), and T (thymine), where A and T are complementary, as are G and C. Similarly, 0 and 1 are complementary in the binary system. Consequently, 00 and 11 are complementary, 01 and 10 are also complementary. There will be 24 kinds of coding schemes if we adopt four bases A, C, G and T to encode 00, 01, 10 and 11. But only 8 of them conform to the Watson-Crick supplementary rules. The encoding rules are showed in Table 1. For an 8-bit gray scale image, each pixel value can be transformed into a binary sequence of length 8. Then, the binary stream is encoded into a DNA sequence of length 4 according to the DNA encoding rules in Table 1, and vice versa.

For instance, the gray scale value of a pixel is "145", "01111110" is the corresponding binary number, a DNA sequence "GCAC" is obtained via DNA encoding rule 1. Inversely, if the DNA sequence is "AGCT", "78" is gained by decoding the DNA sequence with Rule 4.

2.3 DNA sequence insertion and deletion pseudo-operations

The DNA deletion and insertion pseudo-operation is described as a DNA sequence cut from a random position and inserted into somewhere of another DNA sequence.

$$\left[L, I^{'}\right] = \mathrm{DNAcut}(I, c, k), \tag{4}$$

$$I^{'} = \mathrm{DNApast}(C, p, L), \tag{5}$$

where DNAcut represents the DNA deletion operation, I is the initial sequence, c signifies the deletion site, L denotes the deletion sequence, k is the length of L. Similarly, DNApast means the DNA insertion operation, and p is the insertion site, $I^{'}$ represents the recombined sequence by DNA deletion or DNA insertion, as shown in Fig. 1.

3 Specific encryption process

Double gray scale images with the same length and width of M and N are encrypted with each other. The process of encryption algorithm is shown in Fig. 2.

3.1 Generation of key streams

The key streams for the image encryption algorithm are produced by the following steps:

Table 1 DNA coding rules

Number	0	1	2	3	4	5	6	7
DNA coding rules	00—A 01—C 10—G 11—T	00—A 01—G 10—C 11—T	00—C 01—A 10—T 11—G	00—C 01—T 10—A 11—G	00—G 01—A 10—T 11—C	00—G 01—T 10—A 11—C	00—T 01—C 10—G 11—A	00—T 01—G 10—C 11—A

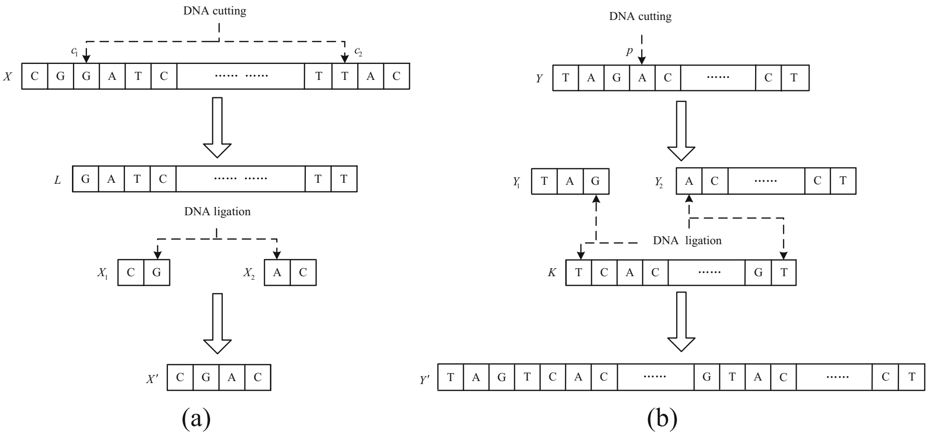

Fig. 1 DNA operations: (**a**) DNA deletion and (**b**) DNA insertion

Step 1: The initial values $x_1 - x_9$ of chaotic sequence are updated as:

$$\begin{cases} x_1{}' = \mathrm{mod}(x_1 + (C_5/C_9), 0.5) \\ x_2{}' = \mathrm{mod}(x_2 + (C_6/C_9), 0.5) \\ x_3{}' = \mathrm{mod}(x_3 + (C_7/C_9), 0.5) \\ x_4{}' = \mathrm{mod}(x_4 + (C_8/C_9), 0.5) \\ x_5{}' = \mathrm{mod}(x_5 + (H_5/H_9), 0.5) \\ x_6{}' = \mathrm{mod}(x_6 + (H_6/H_9), 0.5) \\ x_7{}' = \mathrm{mod}(x_7 + (H_7/H_9), 0.5) \\ x_8{}' = \mathrm{mod}(x_8 + (H_8/H_9), 0.5) \\ x_{ih} = \left[x_1{}', x_2{}', x_3{}', x_4{}', x_5{}', x_6{}', x_7{}', x_8{}', x_1{}'\right] \end{cases}, \tag{6}$$

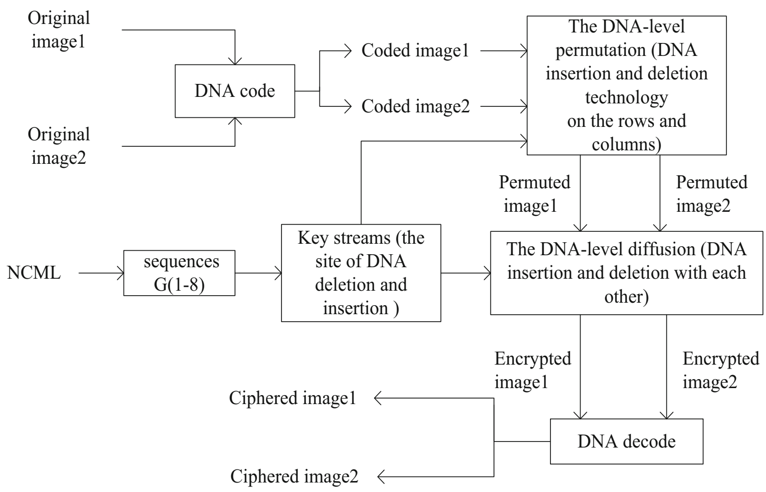

Fig. 2 Encryption architecture

where $C_5 - C_9$ and $H_5 - H_9$ represent the pixels sum of the plain images $C(M, N)$ and $H(M, N)$, respectively, calculated as follows:

$$\begin{cases} C_5 = \sum\limits_{i=M/4+1}^{M/2} \sum\limits_{j=1}^{N} C(i,j) \\ C_6 = \sum\limits_{i=M/4+3}^{M/2+2} \sum\limits_{j=1}^{N} C(i,j) \\ C_7 = \sum\limits_{i=M/4+5}^{M/2+4} \sum\limits_{j=1}^{N} C(i,j) \\ C_8 = \sum\limits_{i=M/4+7}^{M/2+6} \sum\limits_{j=1}^{N} C(i,j) \\ C_9 = \sum\limits_{i=1}^{M} \sum\limits_{j=1}^{N} C(i,j) \end{cases}, \tag{7}$$

$$\begin{cases} H_5 = \sum\limits_{i=1}^{M} \sum\limits_{j=2}^{N/2+1} H(i,j) \\ H_6 = \sum\limits_{i=1}^{M} \sum\limits_{j=2}^{N/2+3} H(i,j) \\ H_7 = \sum\limits_{i=1}^{M} \sum\limits_{j=2}^{N/2+5} H(i,j) \\ H_8 = \sum\limits_{i=1}^{M} \sum\limits_{j=2}^{N/2+7} H(i,j) \\ H_9 = \sum\limits_{i=1}^{M} \sum\limits_{j=1}^{N} H(i,j) \end{cases}, \tag{8}$$

Step 2: The NCML is iterated for $1000 + 8N + M$ times with the eight updated initial values. The first 1000 values are abandoned to avoid the transient effect and the key stream matrix $G(8, M + 8N)$ is obtained. Then the matrix is decomposed into twelve sequences:

$$\begin{array}{l} L_0 = (G(1,1:M) + G(6,1:M)) \times 10^3 \\ L_1 = (G(2,1:M) + G(7,1:M)) \times 10^3 \\ L_2 = (G(3,1:M) + G(8,1:M)) \times 10^3 \\ L_3 = (G(2,M+1:M+4N) + G(5,M+1:M+4N)) \times 10^3 \\ L_4 = (G(3,M+1:M+4N) + G(6,M+1:M+4N)) \times 10^3 \\ L_5 = (G(4,M+1:M+4N) + G(7,M+1:M+4N)) \times 10^3, \\ L_6 = (G(5,1:M) + G_2(2,1:M)) \times 10^3 \\ L_7 = (G(6,1:M) + G_2(3,1:M)) \times 10^3 \\ L_8 = (G(7,1:M) + G_2(4,1:M)) \times 10^3 \\ L_9 = (G(6,M+1:M+4N) + G(1,M+1:M+4N)) \times 10^3 \\ L_{10} = (G(7,M+1:M+4N) + G(2,M+1:M+4N)) \times 10^3 \\ L_{11} = (G(8,M+1:M+4N) + G(3,M+1:M+4N)) \times 10^3 \end{array} \tag{9}$$

Step 3: The key streams kr, kr_1, cr, cr_1, cr_2, pr, pr_1, pr_2 are used to control the locations and the lengths of the insertion and deletion sequences in the row encryption. In a similar way, kc, kc_1, cc, cc_1, cc_2, pc, pc_1, pc_2 are applied to control the locations and the lengths of the DNA insertion and deletion operation of the columns. The key streams are produced by means of arranging the sequences randomly:

$$\begin{cases} kr = \mathrm{mod}\left(\mathrm{floor}\left(L_2(1:20) \times 10^{(3)}\right), 4N\right) + 1 \\ kr_1(i,j) = \mathrm{mod}(\mathrm{floor}(L_0(i) \times L_1(j)), 4N) + 1 \\ cr(i,:) = \mathrm{mod}\left(\mathrm{floor}\left(L_1 \times 10^{(3)}\right), 4N - kr(i) + 1\right) + 1 \\ cr_1(i,j) = \mathrm{mod}(\mathrm{floor}(L_0(i) \times L_2(j)), 4N - kr_1(i,j) + 1) + 1 \\ cr_2(i,j) = \mathrm{mod}(\mathrm{floor}(L_6(i) \times L_8(j)), 4N) + 1 \\ pr = \mathrm{mod}\left(floor\left(L_5 \times 10^{(3)}\right), 4N\right) + 1 \\ pr_1(i,j) = \mathrm{mod}(\mathrm{floor}(L_2(i+10) \times L3(j+30)), 4N - kr_1(i,j) + 1) + 1 \\ pr_2(i,j) = \mathrm{mod}(\mathrm{floor}(L_8(i) \times L_7(j)), 4N) + 1 \end{cases}, \tag{10}$$

$$\begin{cases} kc = \mathrm{mod}\left(\mathrm{floor}\left(L_6(1:20) \times 10^{(3)}\right), M\right) + 1 \\ kc_1(i,j) = \mathrm{mod}(\mathrm{floor}(L_4(i) \times L_{10}(j)), M) + 1 \\ cc = \mathrm{mod}\left(\mathrm{floor}\left(L_5 \times 10^{(3)}\right), M - kc(i) + 1\right) + 1 \\ cc_1(i,j) = \mathrm{mod}(\mathrm{floor}(L_6(i+10) \times L_5(j)), M - kc_1(i,j) + 1) + 1 \\ cc_2(i,j) = \mathrm{mod}(\mathrm{floor}(L_9(i) \times L_{11}(j)), M) + 1 \\ pc = \mathrm{mod}\left(\mathrm{floor}\left(L_9 \times 10^{(3)}\right), M\right) + 1 \\ pc_1(i,j) = \mathrm{mod}(\mathrm{floor}(L_5(i+10) \times L_{11}(j)), M - kc_1(i,j) + 1) + 1 \\ pc_2(i,j) = \mathrm{mod}(\mathrm{floor}(L_6(i+10) \times L_{10}(j)), M) + 1 \end{cases}, \tag{11}$$

The pseudo-random number test standard of NIST SP800–22 (Revision 1a) [29] is utilized to test the characteristics of the key streams. The SP800–22 (Revision 1a) includes 15 kinds of test methods, and each test calculates one or more P values. If the P value is greater than 0.01, then the test passes, otherwise, the test fails. The tested key streams are viewed as random only all tests passed. All the keys have passed the test. To save the space, here we only give the test results of three keys discretionarily. As indicated in Tables 2, 3 and 4, the test results indicate that the keys are random.

3.2 Encryption algorithm

The image encryption process is accomplished by two operations: permutation and diffusion. The positions of pixels are changed in the permutation stage and the values of pixels of the original images are changed by diffusion. The permutation operations: Two original images perform DNA deletion and insert operation on the rows and the columns. The diffusion operations: Two confused images are used as keys each other to carry out DNA deletion and insert operation again. The encryption process is described as follows:

Table 2 Test results of cr_1

Test item	P value	Test result
The frequency (monobit) test	0.8524	Pass
Frequency test within a block	0.2308	Pass
The runs test	0.8634	Pass
Test for the longest-run-of-ones in a block	0.2480	Pass
The binary matrix rank test	0.1180	Pass
The discrete Fourier transform (spectral) test	0.0207	Pass
The non-overlapping template matching test	0.8023	pass
The overlapping template matching test	0.6739	Pass
Maurer's Buniversal statistical test	0.5563	Pass
The linear complexity test	0.2614	Pass
The serial test	{0.8664,0.7763}	Pass
The approximate entropy test	0.5885	Pass
The cumulative sums (Cusums) test	{0.5955,0.9935}	Pass
The random excursions test	{0.2886,0.3398,0.1167,0.0304,0.6728,0.8571,0.5785,0.6370}	Pass
The random excursion variant test	{0.8964,0.7624,0.7866,0.8599,0.6314,0.6447,0.5057,0.2367, 0.3667,0.7327,0.4302,0.5484,0.8392,0.7327,0.8541,0.8870, 0.8255,0.9528}	Pass

Two original images are rearranged

The matrix $D_1(M, N)$ is obtained by carrying out the bitwise exclusive-OR operation for the plaintext matrix $C(M, N)$ and the key matrix $k(M, N)$, similarly, the matrix $D_2(M, N)$ is attained

Table 3 Test results of pr_2

Test item	P value	Test result
The frequency (monobit) test	0.6614	Pass
Frequency test within a block	0.6167	Pass
The runs test	0.5935	Pass
Test for the longest-run-of-ones in a block	0.5264	Pass
The binary matrix rank test	0.0849	Pass
The discrete Fourier transform (spectral) test	0.5326	Pass
The non-overlapping template matching test	0.4375	Pass
The overlapping template matching test	0.8782	Pass
Maurer's Buniversal statistical test	0.0949	Pass
The linear complexity test	0.6615	Pass
The serial test	{0.8734,0.8302}	Pass
The approximate entropy test	0.4012	Pass
The cumulative sums (Cusums) test	{0.9290,0.9161}	Pass
The random excursions test	{0.7273,0.8610,0.2162,0.0597,0.7101,0.7834,0.2189,0.9423}	Pass
The random excursion variant test	{0.5192,0.4926,0.4507,0.5817,0.8572,0.9499,0.9482,0.9204, 0.9384,0.3396,0.0981,0.0459,0.1112,0.1909,0.2598,0.4681, 0.5447,0.4061}	Pass

Table 4 Test results of *kc*

Test item	*P* value	Test result
The frequency (monobit) test	0.6877	Pass
Frequency test within a block	0.8691	Pass
The runs test	0.8276	Pass
Test for the longest-run-of-ones in a block	0.5907	Pass
The binary matrix rank test	0.0235	Pass
The discrete Fourier transform (spectral) test	0.4855	Pass
The non-overlapping template matching test	0.6795	Pass
The overlapping template matching test	0.5464	Pass
Maurer's Buniversal statistical test	0.6492	Pass
The linear complexity test	0.0231	Pass
The serial test	{0.8733,0.9096}	Pass
The approximate entropy test	0.5107	Pass
The cumulative sums (Cusums) test	{0.7908,0.8521}	Pass
The random excursions test	{0.5589,0.6680,0.2064,0.0933,0.1752,0.1197,0.1715,0.5340}	Pass
The random excursion variant test	{0.1719,0.1378,0.1370,0.1967,0.3479,0.5212,0.5575,0.6085, 0.4179,0.2801,0.9822,0.9176,0.5795,0.3681,0.3172,0.2226, 0.3491,0.4265}	Pass

by the other plaintext matrix $H(M, N)$ and the key matrix $k(M, N)$. Then $D_1(M, N)$ and $D_2(M, N)$ are encoded into their associated DNA matrices $E(M, 4N)$ and $F(M, 4N)$ with the coding rules r_1 and r_2, respectively.

$$\begin{cases} k(i,j) = \mathrm{mod}(\mathrm{floor}(L_1(i) \times L_9(j)), 256) \\ r_1 = 5 \\ r_2 = 7 \end{cases}, \tag{12}$$

Permutation operation

The DNA deletion and insert operation is performed on the rows and columns of the matrix $E(M, 4N)$ and the matrix $F(M, 4N)$, respectively.

The matrix $E(M, 4N)$ is scrambled by carrying out the DNA deletion and insert operation with the method below:

Step 1: Row permutation: A DNA sequence Ir of length kr is cut off from the j-th row of the DNA matrix $E(M, 4N)$, and the deletion site is $cr(:, j)$. Next, Ir is inserted into the stickup position $pr(j+1)$ of the $(j+1)$-th row of $E(M, 4N)$, while $j = 1, 2, \ldots, M-1$. When $j = M$, Ir is inserted into the 1st row of the matrix, the above operation is replicated for four times, and the matrix $E_1(M, 4N)$ is generated.

Step 2: Column permutation: Likewise, the columns of $E_1(M, 4N)$ are scrambled by the following procedure:

A DNA sequence Ic of length kc is cut off from the j-th column of the DNA matrix $E_1(M, 4N)$, and the deletion site is $cc(:, j)$. Next, Ic is inserted into the stickup position $pc(j+1)$ of the $(j+$

1)-th column of $E_1(M, 4N)$, while $j = 1, 2, \ldots, 4N-1$. If $j = 4N$, Ic is inserted into the first column of the matrix, and the operation of column-permutation is replicated twice:

$$
\begin{array}{l}
\textit{for } i = 1:2 \\
\quad \textit{for } j = 1:4N \\
\qquad [Ic, E_1\{1,j\}] = \mathrm{DNAcutc}(E_1\{1,j\}, cc(i,j), kc(i)); \\
\qquad \textit{if } j < 4N \\
\qquad\quad E_1\{1,j+1\} = \mathrm{DNApastc}(E_1\{1,j+1\}, pc(j+1), Ic); \\
\qquad \textit{else} \\
\qquad\quad E_1\{1,1\} = \mathrm{DNApastc}(E_1\{1,1\}, v(i), Ic); \\
\qquad \textit{end} \\
\quad \textit{end} \\
\textit{end}
\end{array}
$$

Step 3: The same operations of the matrix $E(M, 4N)$ are executed on the matrix $F(M, 4N)$. Thus the shuffled DNA matrices $E_2(M, 4N)$ and $F_2(M, 4N)$ are obtained.

Diffusion operation

DNA operation is executed on the confused image $E_2(M, 4N)$ $((F_2(M, 4N)))$ with the key $F_2(M, 4N)$ $(E_2(M, 4N))$. Similarly, the diffusion operation is divided into two steps: row diffusion and column diffusion:

Step 1: Row diffusion: Firstly, a DNA sequence Ix of length $kr_1(:,j)$ is clipped from the j-th row of the matrix $E_2(M, 4N)$, and the deletion site is $cr_1(:,j)$. Then Ix is pasted into the insertion site $pr_2(:,j+1)$ of the $(j+1)$-th row of the matrix $F_2(M, 4N)$, while $j = 1, 2, \ldots, M-1$. When $j = M$, Ix is pasted into the stickup position $pr_2(:, 1)$ of the 1st row of $F_2(M, 4N)$.

Secondly, a DNA sequence Hx of length $kr_1(:,j-1)$ is sheared off from the deletion site $cr_2(:, j)$ of the matrix $F_2(j, :)$. Then Hx is inserted into the stickup position $pr_1(:,j-1)$ of the $(j-1)$-th row of the matrix $E_2(M, 4N)$, while $j = 2, 3, \ldots, M$. When $j = 1$, a DNA sequence Hx of length $kr_1(:, M)$ is cut off from the first row of the matrix $F_2(M, 4N)$, and inserted into the insertion site $pr_1(:, M)$ of the M-th row of $E_2(M, 4N)$. The above operations are recycled for two times, and then the matrix $E_3(M, 4N)$ and the matrix $F_3(M, 4N)$ are gained, which can be outlined as:

$$
\begin{array}{l}
\textit{for } i = 1:2 \\
\quad \textit{for } j = 1:M \\
\qquad [Ix, E_2\{j,1\}] = \mathrm{DNAcutr}(E_2\{j,1\}, cr_1(i,j), kr_1(i,j)); \\
\qquad \textit{if } j < M \\
\qquad F_2\{j+1,1\} = \textit{DNApastr}(F_2\{j+1,1\}, pr_2(i,j+1), Ix); \\
\qquad \textit{else} \\
\qquad F_2\{1,1\} = \mathrm{DNApastr}(F_2\{1,1\}, pr_2(i,1), Ix); \\
\qquad \textit{end} \\
\quad \textit{end} \\
\textit{for } j = 2:M \\
\quad [Hx, F_2\{j,1\}] = \textit{DNAcutr}(F_2\{j,1\}, cr_2(i,j), kr_1(i,j-1)); \\
\quad E_2\{j-1,1\} = \mathrm{DNApastr}(E_2\{j-1,1\}, pr_1(i,j-1), Hx); \\
\textit{end} \\
\quad [Hx, F_2\{1,1\}] = \textit{DNAcutr}(F_2\{1,1\}, cr_2(i,1), kr_1(i,M)); \\
\quad E_2\{M,1\} = \textit{DNApastr}(E_2\{M,1\}, pr_1(i,M), Hx); \\
\textit{end}
\end{array}
$$

Step 2: Column diffusion: Likewise, the columns of $E_3(M, 4N)$ and $F_3(M, 4N)$ are scrambled as follows:

Firstly, a DNA sequence Iy of length $kc_1(:, j)$ is clipped from the j-th column of the matrix $F_3(M, 4N)$, and the deletion site is $cc_1(:, j)$. Then Iy is pasted into the insertion site $pc_2(:, j+1)$ of the $(j+1)$-th column of the matrix $E_3(M, 4N)$, while $j = 1, 2, \ldots, 4N-1$. If $j = 4N$, Iy is pasted into the stickup position $pc_2(:, 1)$ of the 1st column of $E_3(M, 4N)$.

Secondly, a DNA sequence Hy of length $kc_1(:, j-1)$ is sheared off from the deletion site $cc_2(:, j)$ of the matrix $E_3(M, 4N)$. Then Hy is inserted into the stickup position $pc_1(:, j-1)$ of the $(j-1)$-th column of the matrix $F_3(M, 4N)$, while $j = 2, 3, \ldots, 4N$. When $j = 1$, a DNA sequence Hy of length $kc_1(:, 4N)$ is cut off from the 1st column of the matrix $E_3(M, 4N)$, and inserted into the insertion site $pc_1(:, 4N)$ of the $4N$-th column of $F_3(M, 4N)$. The above operations are recycled for two times.

Generation of cipher-images

Two encryption images are obtained by encoding the two encryption matrices, where the DNA coding rules are r_3 and r_4, respectively.

$$r_3 = 2, r_4 = 6 \tag{13}$$

3.3 Decryption algorithm

The decryption algorithm is the inverse process of the encryption algorithm. The decryption process is shown in Fig. 3, and the steps are summarized as follows:

Step 1: The arguments are calculated by the method described in Section 3.1. With the correct arguments, the chaotic system is iterated to generate the pseudo-random sequences, the key streams are obtained via the formulas (9) and (10).

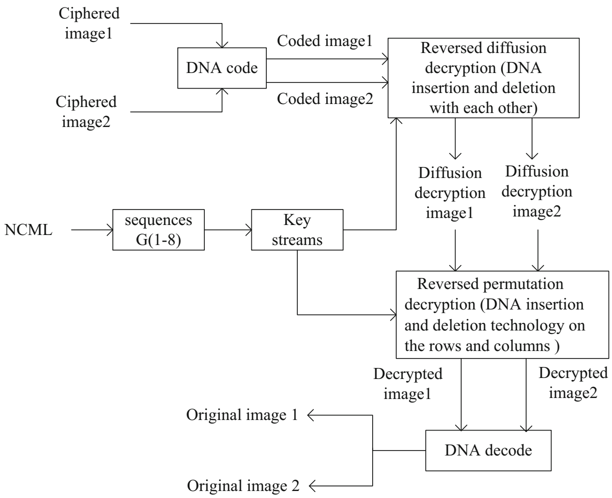

Fig. 3 Decryption architecture

Step 2: The encryption images are encoded by the encoding rules r_3 and r_4, and the two DNA matrices $E_4(M, 4N)$ and $F_4(M, 4N)$ are obtained.
Step 3: $E_4(M, 4N)$ and $F_4(M, 4N)$ are used as keys of each other to carry out DNA operation on their rows and columns.

Reversed column diffusion: DNA deletion operation is administered on the site $pc_1(:, j)$ of the j-th column of the matrix $F_4(M, 4N)$, and a DNA sequence hy of length $kc_1(:, j)$ is cut off from the j-th column of the matrix $F_4(M, 4N)$. Then hy is pasted into the insertion site $cc_2(:, j+1)$ of the $(j+1)$-th column of the matrix $E_4(M, 4N)$, while $j = 1, 2, \ldots, 4N-1$. When $j = 4N$, hy is pasted into the stickup position $cc_2(i, 1)$ of the first column of $E_4(M, 4N)$.

A DNA sequence hy of length $kc_1(i, j-1)$ is sheared off from the deletion site $pc_2(:, j)$ of the matrix $E_4(M, 4N)$. Then Hy is inserted into the stickup position $cc_1(i, j-1)$ of the $(j-1)$-th column of the matrix $F_4(M, 4N)$, while $j = 4N, 4N-1, ..., 2$. If $j = 1$, a DNA sequence hy of length $kc_1(:, 4N)$ is cut off from the 1st column of the matrix $E_4(M, 4N)$, and inserted into the insertion site $cc_1(:, 4N)$ of the $4N$-th column of $F_4(M, 4N)$. The above operations are recycled for two times.

Reversed row permutation: The method to remove the effect of the row diffusion is similar to that of the reversed column diffusion.

```
for i = 2 : −1 : 1
  for j = 1 : M
    [Ix, E3{j, 1}] = DNAcutr(E3{j, 1}, pr1(i, j), kr1(i, j));
    if j < M
    F3{j + 1, 1} = DNApastr(F3{j + 1, 1}, cr2(i, j + 1), Ix);
     else
    F3{1, 1} = DNApastr(F3{1, 1}, cr2(i, 1), Ix);
    end
  end
  for j = 2 : M
     [Ix, F3{j, 1}] = DNAcutr(F3{j, 1}, pr2(i, j), kr1(i, j−1));
      E3{j−1, 1} = DNApastr(E3{j−1, 1}, cr1(i, j−1), Ix);
  end
  [Ix, F3{1, 1}] = DNAcutr(F3{1, 1}, pr2(i, 1), kr1(i, M));
  E3{M, 1} = DNApastr(E3{M, 1}, cr1(i, M), Ix);
end
```

Then the DNA matrices $E_2(M, 4N)$ and $F_2(M, 4N)$ are gained.

Step 4: $E(M, 4N)$ and $F(M, 4N)$ are configured by performing the inversed operations corresponding to the permutation operations in the encryption method.
Step 5: $E(M, 4N)$ and $F(M, 4N)$ are decoded into the corresponding binary matrices $D_1(M, N)$ and $D_2(M, N)$ with the rules r_1 and r_2.
Step 6: After implementing the bitwise exclusive-OR operation for $D_1(M, N)$ and $k(M, N)$, the decryption image $C(M, N)$ is obtained. Likewise, the decryption image $H(M, N)$ is obtained after implementing the bitwise exclusive-OR operation for $D_2(M, N)$ and $k(M, N)$.

4 Simulation and exhaustive analysis

The experiments are carried out in a computer of Core i5 CPU 4.0 GHz, the standard grey 256×256 images ‘Lena’, ‘Camera’ and ‘Dollar’, ‘Airfield’ are employed as the test images, shown in Fig. 4(a)–(d). All experiments are manipulated with Matlab 2012a on a PC with windows 7 64-bit operating system. The encryption images are shown in Fig. 4(e)–(h), and the decryption images are shown in Fig. 4(i)–(l). From Fig. 4, it can be seen the proposed image cryptosystem could encrypt the plaintext images into the noise-like ciphered images without any visual information leakage. Meanwhile, the decrypted images are identical to the original plaintext images.

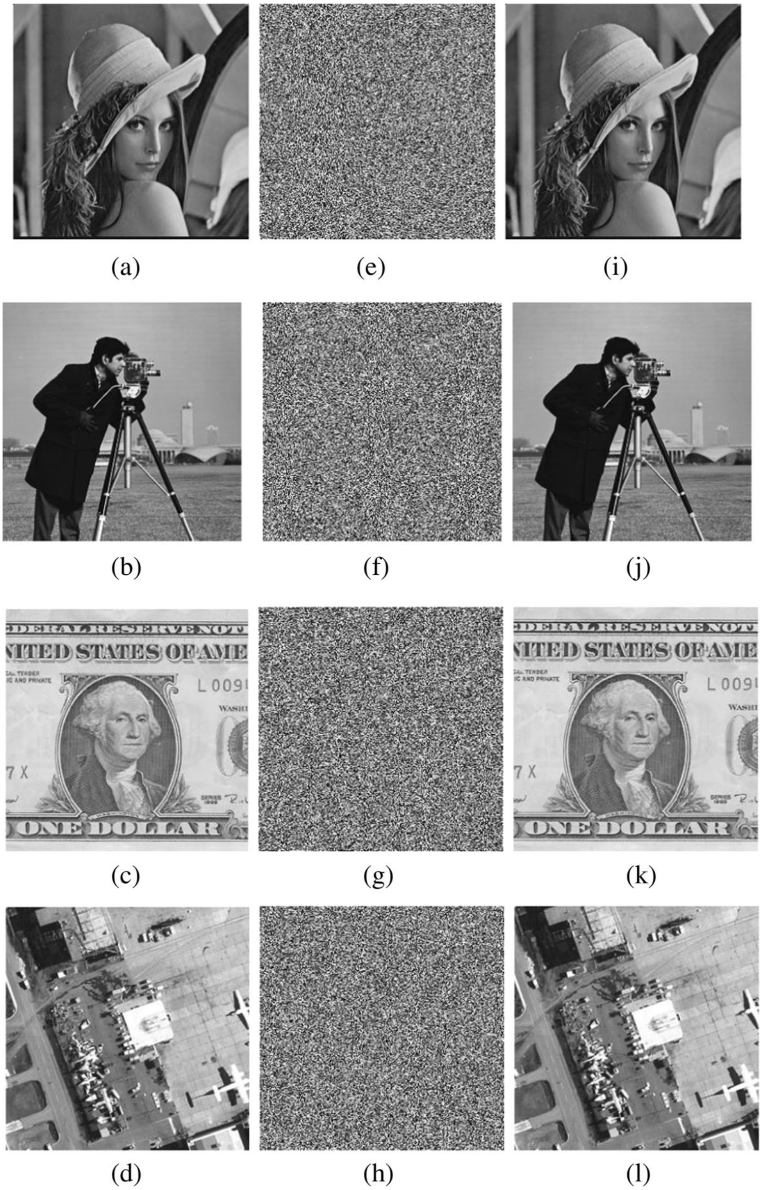

Fig. 4 Experimental results: (**a**)–(**d**) original image 'Lena' 'Camera' and 'Airfield' 'Dollar'. (**e**)–(**h**) encryption image of (**a**)–(**d**), respectively. (**i**)–(**l**) decryption image of (**e**)–(**h**), respectively

4.1 Key space

The ability to resist the brute-force attack is up to the size of key space. In addition to the control parameters (e, u) and the initial conditions ($x_1 - x_8$), the sum of all and partial pixels of the original images ($C_5 - C_9, H_5 - H_9$) are also the secret keys of the encryption algorithm. If the computation precision of the control parameters and the initial conditions is 10^{-15}, then the key space size of our image encryption algorithm will be greater than $((10^{15})^{10}) \approx 2^{500}$, which is large enough to make the brute-force attack ineffective.

4.2 Sensitivity analysis

4.2.1 Key sensitivity

A secure image cryptosystem should be sensitive to the slightest modifications of the key to guarantee the brute-force attack infeasibility. To test the sensitivity of our encryption algorithm, the wrong keys slightly different from the encryption ones are fetched to decrypt the ciphered image. The qualitative and the quantitative methods are introduced to measure the difference between two images. As it can be seen from Fig. 5, the differences between the decryption images are difficult to distinguish by visual inspection. The number of pixels change rate (NPCR) and the unified average changing intensity (UACI) are used. The NPCR represents the percentage of different pixels between two decrypted images with a tiny difference of the decryption key. The UACI represents the average intensity of differences between the two decrypted images.

$$\mathrm{NPCR} = \frac{\sum_{i=1}^{M} \sum_{j=1}^{N} |\mathrm{Sign}(C_1(i,j) - C_2(i,j))| \times 100\%}{M \times N}, \tag{14}$$

$$\mathrm{UACI} = \frac{1}{M \times N} \left[\sum_{i,j} \frac{|C_1(i,j) - C_2(i,j)|}{255} \right] \times 100\%, \tag{15}$$

$$\mathrm{Sign}(x) = \begin{cases} 1, x > 0; \\ 0, x = 0; \\ -1, x < 0. \end{cases}, \tag{16}$$

where M and N mean the width and the height of the decrypted image, C_1 and C_2 represent the decrypted image before and after one key is changed. The expected values of PSNR and UACI for the given ‘Lena’, ‘Camera’, ‘Dollar’, ‘Airfield’ images and random images are 99.6094% and 30.7187%, 99.6094% and 31.1691%, 99.6094% and 31.8672%, 99.6094% and 30.6862%, respectively [29]. It can be seen from Table 5 that the calculated results of NPCR and UACI are very close to the expected values, and it indicates that the proposed image encryption algorithm has excellent key sensitivity and is sufficient to resist the differential attack.

4.2.2 Differential attacks analysis

For an encryption algorithm, the ability of resisting differential attack is connected with the sensitivity to plaintext image and ciphertext image.

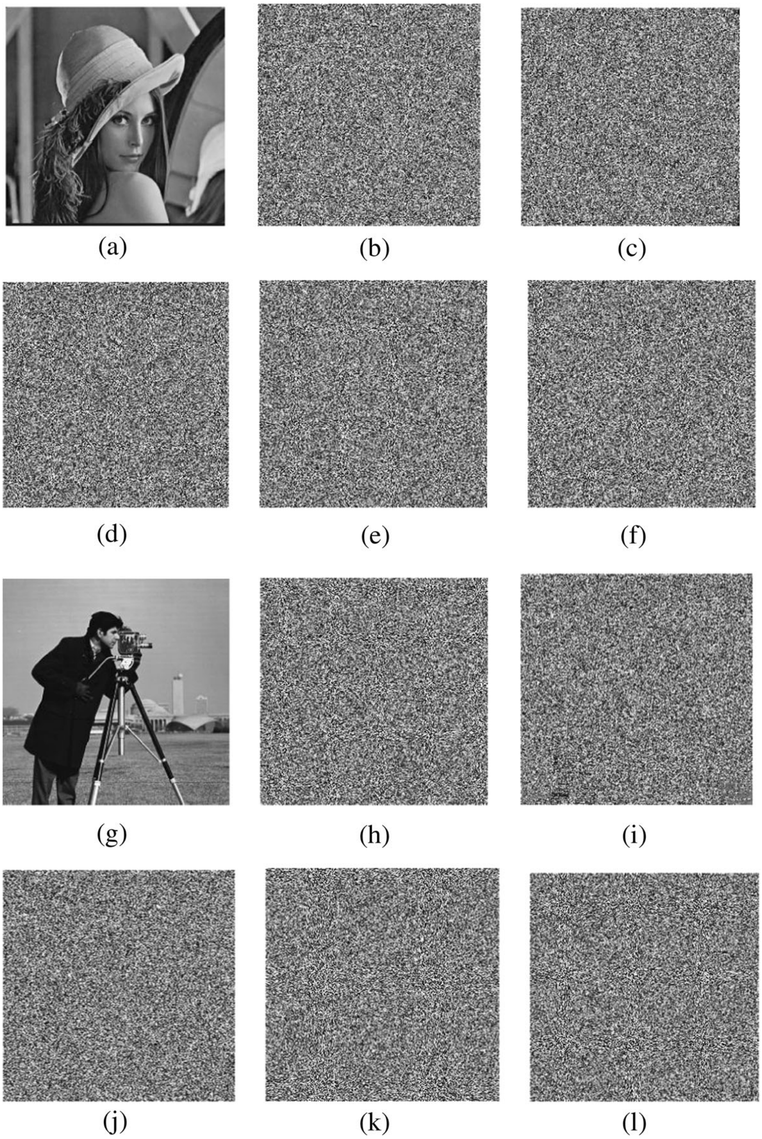

Fig. 5 Key sensitivity tests: (**a**) and (**g**) are the plaintext images, decryption images while (**b**) and (**h**) $x_1 + 10^{-15}$, (**c**) and (**i**) $x_4 + 10^{-15}$, (**d**) and (**j**) $x_7 + 10^{-15}$, (**e**) and (**k**) $u + 10^{-15}$, (**f**) and (**l**) $e + 10^{-15}$

Plaintext sensitivity analysis: A secure image encryption algorithm is highly sensitive to any change in the plaintext image, which means that one-pixel change will generate a completely different encryption image. Similarly, the standards of NPCR and UACI are usually used to examine the performance of resisting the known/chosen plaintext attacks. The expected values of NPCR and UACI for an 8-bit grey image are 99.6094% and 33.4635%, respectively.

$$\mathrm{NPCR}_{expected} = (1 - 1/L) \times 100\%, \tag{17}$$

Table 5 Key sensitivity

Key change value		Lena	Camera	Dollar	Airfield
$x_1 \times 10^{-15}$	NPCR	99.5939%	99.5682%	99.6277%	99.6205%
	UACI	30.7560%	31.1381%	31.8348%	30.6837%
$x_4 \times 10^{-15}$	NPCR	99.5935%	99.5677%	99.5781%	99.5902%
	UACI	30.7581%	31.1350%	31.2597%	30.3024%
$x_7 \times 10^{-15}$	NPCR	99.5940%	99.5680%	99.6269%	99.6153%
	UACI	30.7566%	31.1378%	31.8363%	30.3656%
$u \times 10^{-15}$	NPCR	99.5936%	99.5677%	99.6323%	99.6170%
	UACI	30.7534%	31.1359%	32.0471%	30.6378%
$e \times 10^{-15}$	NPCR	99.5937%	99.5677%	99.6170%	99.6201%
	UACI	30.7547%	31.1350%	31.8251%	30.6651%

$$\mathrm{UACI}_{expected} = \frac{1}{L^2}\left[\sum_{1}^{L-1} L(L+1)/(L-1)\right] \times 100\%, \tag{18}$$

where L is the grey level of the plaintext image.

Ciphertext sensitivity analysis: after minor difference of the ciphertext, the image restored by the decryption system is different from the plaintext image. The analysis is described as follows:

(1) Assume that the plain image C was encrypted by the proposed scheme with secret key K to generate the ciphered image, denoted as C_1.
(2) Change C_1 by only one pixel to get the image named by C_2, then C_2 is decrypted with the same key to get the recovered image, denoted as C_3.
(3) Compare the difference between C and C_3.

The results of the plaintext and the ciphertext sensitivity analysis are enumerated in Tables 6 and 7. It can be seen that the results of NPCR and UACI are fairly close to the theoretical values. Hence, the introduced image encryption algorithm is enough secure against differential attacks.

4.3 Statistical analysis

4.3.1 Correlation coefficient

If the image has a large correlation between adjacent pixels, the attacker can obtain relevant information of the original image by analyzing the correlation between pixels.

Table 6 Plaintext sensitivity analysis

Image	NPCR (%)		UACI (%)	
	Average value	Expected value	Average value	Expected value
Lena	99.6204	99.6094	30.7972	30.7187
Camera	99.6307	99.6094	31.2802	31.1691
Dollar	99.5987	99.6094	31.9509	31.8672
Airfield	99.5941	99.6094	30.7034	30.6862

Table 7 Ciphertext sensitivity analysis

Image	NPCR (%)		UACI (%)	
	Average value	Expected value	Average value	Expected value
Lena	99.6204	99.6094	30.7972	30.7187
Camera	99.6307	99.6094	31.2802	31.1691
Dollar	99.5987	99.6094	31.9509	31.8672
Airfield	99.5941	99.6094	30.7034	30.6862

Therefore, the closer the correlation coefficient among adjacent pixels of the ciphered image is to 0, the more secure the ciphered image is. To test the correlation of plaintext image and ciphered image, 10,000 pairs of two adjacent pixels are selected randomly from original image and ciphered image, respectively. The correlation among pixels is computed by:

$$E(x) = \frac{1}{N}\sum_{i=1}^{N} x_i, \tag{19}$$

$$D(x) = \frac{1}{N}\sum_{i=1}^{N} (x_i - E(x))^2, \tag{20}$$

$$\mathrm{cov}(x, y) = \frac{1}{N}\sum_{i=1}^{N} (x_i - E(x))(y_i - E(y)), \tag{21}$$

$$\gamma_{xy} = \frac{\mathrm{cov}(x, y)}{\sqrt{D(x)}\sqrt{D(y)}}, \tag{22}$$

where x and y are the grey values of two adjacent pixels, $E(x)$ and $D(x)$ signify the expectation and the variance of x and y, respectively, and $\mathrm{cov}(x, y)$ is covariance. The correlation coefficients and the correlation distributions of the plaintext images and the ciphered images are shown in Table 8 and Fig. 6, respectively. Both Table 8 and Fig. 6 reveal that the correlation between two adjacent pixels in the original image is significantly weakened, which indicates the proposed image encryption algorithm could withstand the statistical analysis attack.

4.3.2 Histogram analysis

The pixel statistics of the original image can be acquired by attackers by analyzing the histogram distribution of encryption images. Thus, the histogram distribution of the image would become uniform after the image disposed by an ideal cryptosystem. The histograms of the original image and the encryption image are shown in Fig. 7, where (a)-(h) are correlation distributions of the original and the encryption images 'Lena', and (i)-(p) are the correlation distributions of the original and the encryption

Table 8 Correlation coefficient of plaintext image and ciphered image

Image	Plain image				Ciphered image			
	Horizontal direction	Vertica direction	Diagonal direction	Counter-diagonal direction	Horizontal direction	Vertical direction	Diagonal direction	Counter-diagonal direction
Lena	0.9649	0.9442	0.9182	0.9373	−0.00002	−0.0040	0.0026	0.0009
Camera	0.9570	0.9336	0.9134	0.9183	−0.0074	0.0087	0.0053	0.0081
dollar	0.8256	0.8616	0.7202	0.7188	0.0037	0.0066	−0.0007	0.0089
Airfield	0.9227	0.9142	0.8645	0.8721	−0.0067	0.0025	0.0037	−0.0086
Finger	0.6425	0.5553	0.5113	0.4054	−0.0066	0.0037	0.0131	−0.0043
Aerial	0.7859	0.8341	0.7059	0.7034	−0.0088	−0.0066	−0.0102	−0.0001

images 'Camera'. It is clearly shown that the histograms of the encryption images carry a uniform distribution after performing one round of encryption. χ^2 statistic is usually used to validate the difference between the histograms of the original and the encryption images. For an 8-bit grayscale image of size $M \times N$, χ^2 statistic is described as:

$$\chi^2 = \sum_{i=0}^{255} \frac{(fi - g)^2}{g}, \tag{23}$$

where $g = MN/256$, given the significant level $\alpha = 0.05$, such as:

$$\rho\left\{\chi^2 \geq \chi^2_{0.05}(255)\right\} = 0.05, \tag{24}$$

If $\chi^2 < \chi^2_{0.05}(255) = 293.2478$, we will accept the assumption that the histogram is assumed to be distributed uniformly. As can be seen from Table 9, χ^2 statistics of plain images are much larger than $\chi^2_{0.05}(255)$, while χ^2 statistics of ciphered images is smaller than $\chi^2_{0.05}(255)$.

4.4 Information entropy

Information entropy is one of the most important features of randomness. The more the value of information entropy is close to 8bits, the better the image encryption algorithm resists effectively the known-plaintext and the chosen-plaintext attacks. The information entropy is described below:

$$H(s) = \sum_{i=0}^{2^N - 1} P(s_i) \log_2 \frac{1}{P(s_i)}, \tag{25}$$

where $P(s_i)$ expresses the probability of s_i, s_i is the i-th grey value for $2^N - 1$ level grey image. The information entropies of several original images and the corresponding ciphered images are configured respectively in Table 10.

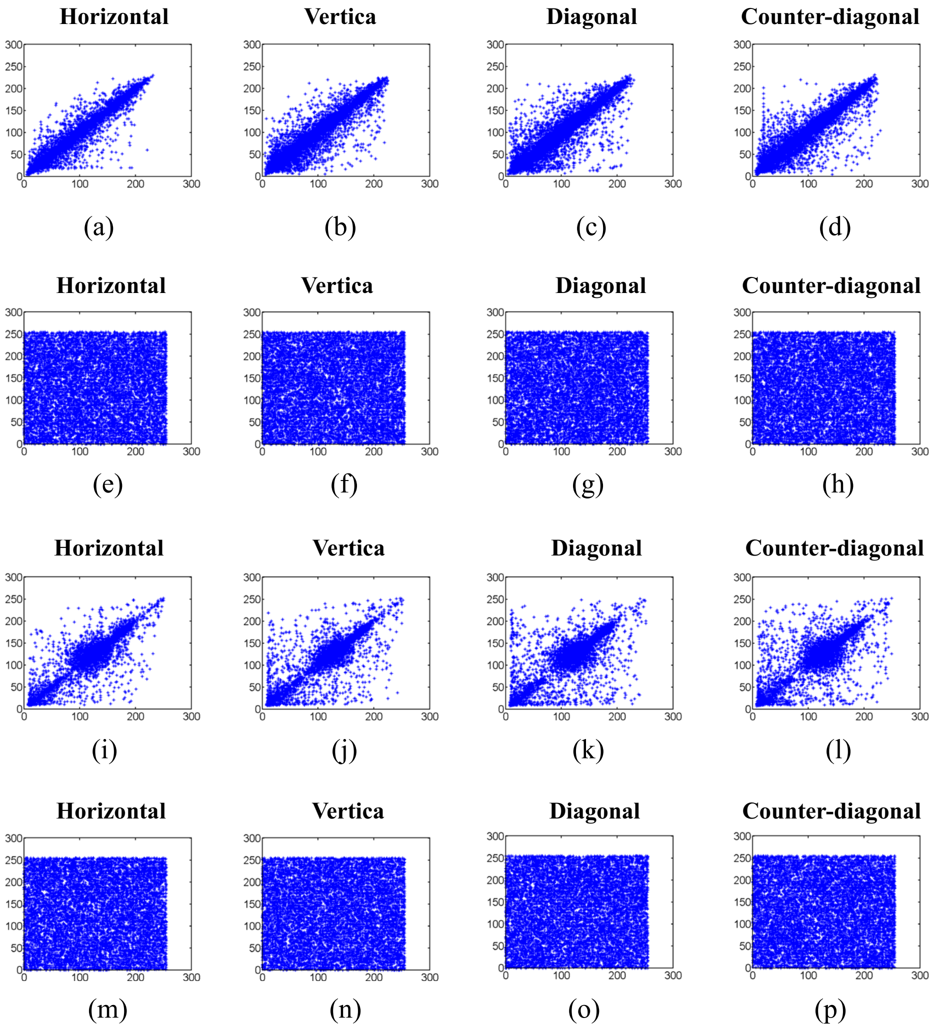

Fig. 6 (**a**)–(**h**) are the original and the encryption images' correlation distributions of 'Lena', (**i**)–(**p**) are the original and the encryption images' correlation distributions of 'Camera'

4.5 Occlusion attack

It will be difficult to decrypt the original image if the data is lost during transmission. A loss manipulation known as occlusion occurs frequently in practical applications. Data blocks of size $128 \times 64, 128 \times 128, 256 \times 128$ are sheared from the encryption

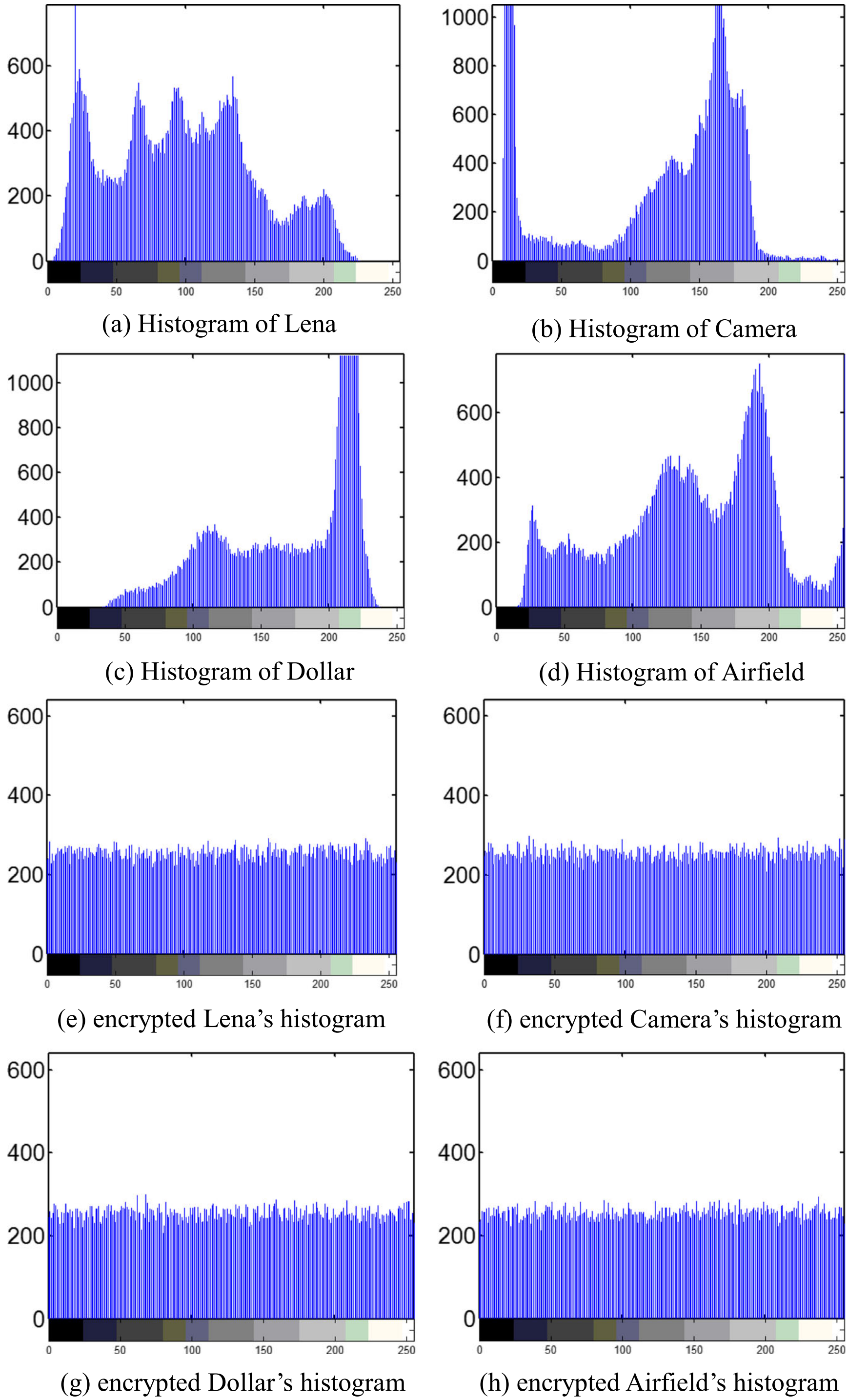

Fig. 7 Histograms of the original images and the encryption images

Table 9 χ^2 statistics of plaintext images and ciphered ones

Image	Lena	Camera	Dollar	Airfield	Finger	Aerial
Plain image	3.3857×10^4	1.1097×10^5	1.3713×10^5	3.1871×10^4	7.2970×10^4	1.1570×10^5
Ciphered image	226.3047	238.8828	246.6875	199.2031	217.4822	229.8984

image to test the robustness of the image encryption algorithm. Figure 8(a)–(c) and (g)–(i), (d)–(f) and (j)–(l) present the corresponding decryption images, which are fuzzy but recognizable. The Peak Signal-to-Noise Ratio (PSNR) between original image and decryption image are computed to analyze the quality of the recovered image after attacking. The larger the PSNR is, the stronger the robustness of the encryption algorithm is.

$$\mathrm{PSNR} = 10 \times \lg \frac{{\mathrm{MAX_I}}^2}{\mathrm{MSE}}, \tag{26}$$

$$\mathrm{MSE} = \frac{1}{M \times N} \sum_{i=1}^{M} \sum_{j=1}^{N} [P(i,j) - D(i,j)]^2, \tag{27}$$

where M, N signify the length and the width of the image, $P(i,j)$ and $D(i,j)$ represent original image and decryption one, respectively, $\mathrm{MAX_I}$ denotes the maximum pixel value in the image. The PSNR comparisons among our algorithm and those in [8, 9, 19, 31] are displayed in Table 11.

4.6 Noise attack

In practical situations, the ciphered images are inevitably affected by all types of noises during transmission. Noise addition will result in degradation and distortion of the image, which consequently influences the decryption effect. To test the noise immunity of our image encryption algorithm, Salt and Pepper noise with different noise densities, i.e., 0.001, 0.005, 0.01 and 0.1, are added into the ciphered images. The test results are visualized in Fig. 9. The PSNR comparisons among our algorithm and others [8, 9, 19, 31] are compiled in Table 12. The comparisons demonstrate the robustness of the proposed image encryption scheme against Salt and Pepper noise is better.

4.7 Encryption speed

An effective image encryption algorithm should run fast enough to meet the requirement of real-time image communication. In terms of the encryption speed, our

Table 10 Information entropies of plaintext images and ciphered images

Image	Lena	Camera	Dollar	Airfield	Finger	Aerial
Size	256×256	256×256	256×256	256×256	256×256	256×256
Plain image	7.5536	7.0097	6.9982	7.6406	7.1075	6.9277
Ciphered image	7.9972	7.9973	7.9973	7.9978	7.9976	7.9975

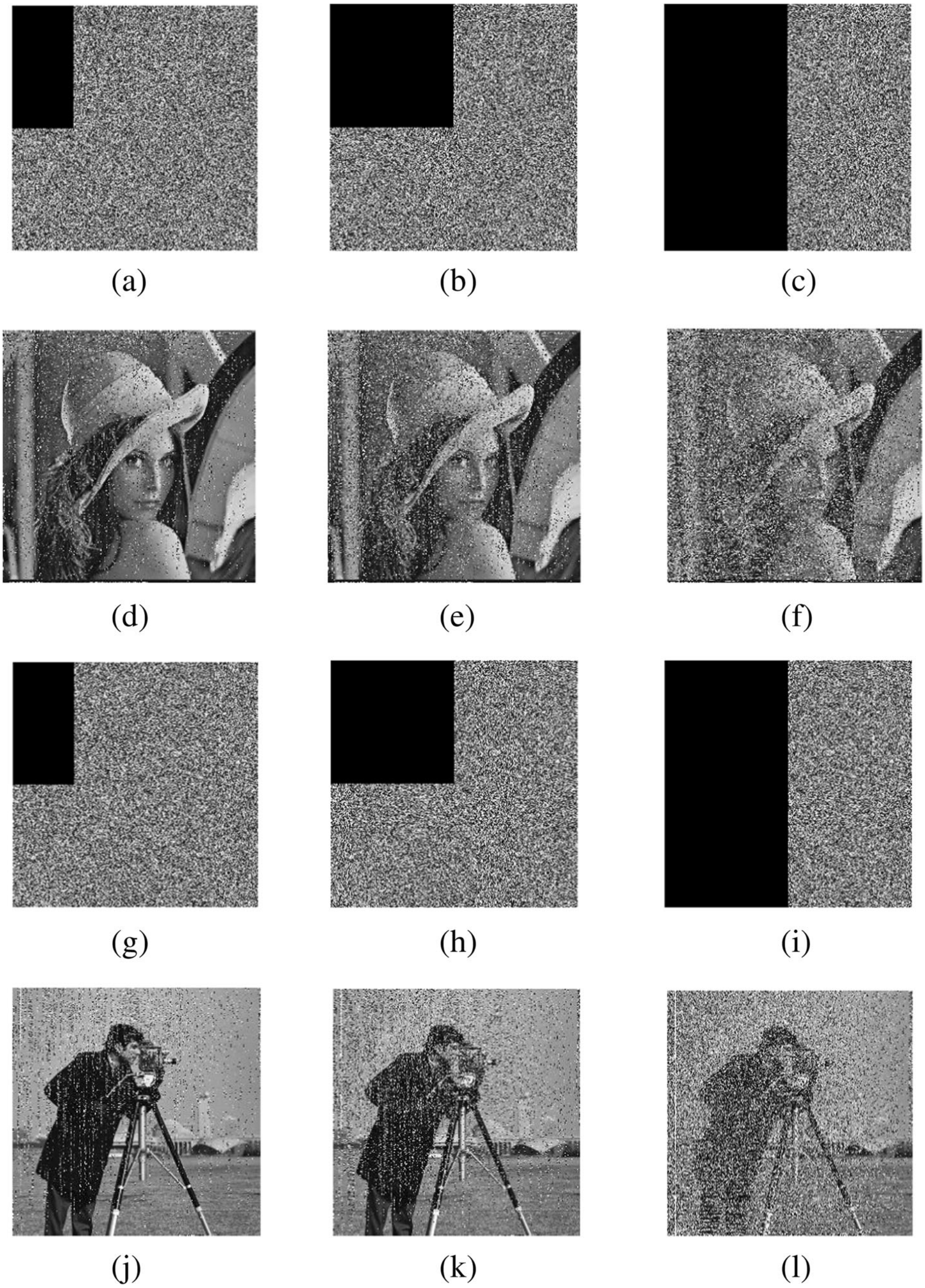

Fig. 8 Encryption images 'Lena' with (**a**) $\frac{1}{8}$; (**b**) $\frac{1}{4}$; (**c**) $\frac{1}{2}$ data occlusion, corresponding decryption images (**d**)–(**f**) from (**a**)–(**c**). Encryption images 'Camera' with (**g**) $\frac{1}{8}$; (**h**) $\frac{1}{4}$; (**i**) $\frac{1}{2}$ data occlusion; corresponding decryption images (**j**)–(**l**) from (**g**)–(**i**)

proposed scheme and other encryption schemes [8, 9, 19, 31] are simulated under the same conditions, as shown in Table 13. It is shown the proposed image encryption algorithm is the fastest than others.

Table 11 PSNR between original image and decryption image under occlusion attack

	Algorithm		Occlusion		
			1/8	1/4	1/2
PSNR (dB)	Ours	Lena	17.5998	14.3349	11.4402
		Camera	17.4488	14.5293	11.3254
	Ref. [19]	Lena	17.0023	14.3316	11.3735
	Ref. [31]	Lena	17.2537	14.3248	11.4208
	Ref. [8]	Lena	17.4013	14.3939	11.3895
	Ref. [9]	Lena	17.6826	14.734	11.4238

4.8 Comparison

The proposed image encryption scheme is compared with the existing image encryption techniques, all statistical results in Tables 11, 12, 13 and 14 are based on images 'Lena' and 'Camera' of size 256 × 256. As shown in Table 14, the key space of our algorithm is the largest one, so the proposed cryptosystem could resist the brute-force attack better. And the proposed image encryption scheme achieves a stably low correlation among adjacent pixels in the decryption images compared with others' schemes. The values of NPCR and UACI in the described image encryption algorithm are closer to the ideal values 99.6094% and 33.4635%, respectively, which outperforms the other image encryption schemes [8, 9, 19, 31, 35]. The information entropy is close to 8 bits, although our scheme does not perform the best. Tables 11 and 12 show that the introduced image encryption algorithm has a better performance in resisting the occlusion and the noise attacks. As shown in Table 13, the speed of our image encryption algorithm is fastest than other algorithms. It indicates that the properties of the described image encryption scheme are acceptable. Hence, the proposed image encryption algorithm is feasible and effective.

Fig. 9 Decryption images under Salt and Pepper noises with (**a**), (**e**) 0.001; (**b**), (**f**) 0.005; (**c**), (**g**) 0.01; and (**d**), (**h**) 0.05 density contamination

Table 12 PSNR between original image and decryption image under Salt and Pepper noise attack

	Algorithm		Density of Salt and Pepper noise			
			0.001	0.005	0.01	0.05
PSNR(dB)	Ours	Lena	38.4028	30.5231	28.3143	21.4101
		Camera	38.2490	31.4498	28.8060	21.5332
	Ref. [19]	Lena	35.5809	28.5361	24.9837	18.1529
	Ref. [31]	Lena	34.801	27.8651	24.8026	18.1102
	Ref. [8]	Lena	39.1247	31.7666	28.2636	21.2504
	Ref. [9]	Lena	38.6725	31.6216	28.2277	21.4003

Table 13 Encryption speed comparison

Scheme	Encryption rate (Mbit/s)
Ours	7.8692
Ref. [19]	1.1787
Ref. [31]	1.3785
Ref. [8]	3.9367
Ref. [9]	2.6973

5 Conclusions

A novel double-image encryption algorithm established on the spatiotemporal chaotic system and DNA technique is designed. The NCML is employed in the encryption algorithm to produce more random sequences, and the test standard of SP800–22 Revision 1a is used to verify the statistical characteristics of pseudo-random sequence. The key streams for the image encryption are generated based on the secret keys and the original images so that the proposed image encryption algorithm could resist the plaintext attacks. DNA-level confusion, DNA-level diffusion and pixel-level diffusion are applied to enhance the security of the proposed scheme. Besides, compared with the existing double-image encryption schemes, the proposed image encryption scheme offers excellent robustness and high encryption speed, which makes the encryption scheme more suitable for real-time applications. Simulation and theoretical analyses show that the effectiveness of double-image encryption scheme. The proposed double-image encryption scheme is helpful to develop multi-image encryption scheme in the future DNA computer and achieve high-speed encryption/decryption and high-level security under the real-time embedded system.

Table 14 Comparison with other algorithms

Scheme	Ours		Ref. [19]	Ref. [31]	Ref. [8]	Ref. [9]	Ref. [35]
	Lena	Camera	Lena	Lena	Lena	Lena	Lena
Key space	2^{500}		2^{182}	2^{400}	2^{349}	2^{249}	2^{233}
Information entropy (bits)	7.9972	7.9973	7.9897	7.9921	7.9975	7.9975	7.9993
NPCR (%)	99.6070	99.6071	99.5911	99.6446	99.5941	99.6246	99.6000
UACI (%)	33.4855	33.4907	33.4798	33.4286	33.3970	33.57774	33.4398

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